Michigan Tech

Lagrange Points and Space Telescopes

Earth–Moon stability and observatory locations

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MA4535 — Nonlinear Dynamics & Chaos | Lecture 14 | 12 October 2026

Outline

1. Locate the five Earth–Moon equilibria.
2. Explain the stability of L_4 and L_5.
3. Identify the systems used by space observatories.
4. Review the mission timeline and station keeping.

Equilibria in the rotating frame

- Recall the scaled rotating equations for the probe at (x, y).

$$x'' - 2y' = \Omega_x \qquad y'' + 2x' = \Omega_y$$

- Define the effective potential function

$$\Omega(x, y) = \frac{1}{2}(x^2 + y^2) + \frac{1-\eta}{\rho_1} + \frac{\eta}{\rho_2}$$

$$\rho_1^2 = (x+\eta)^2 + y^2 \qquad \rho_2^2 = (x-1+\eta)^2 + y^2$$

- A **Lagrange Point** has $x' = y' = x'' = y'' = 0$, so $\nabla\Omega = \mathbf{0}$ [12].

Earth–Moon parameters

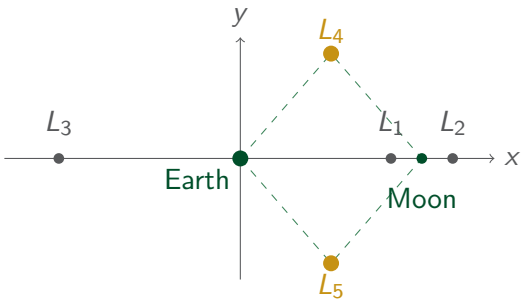
- Earth is primary 1 and the Moon is primary 2.

$$m_E = 5.9722 \times 10^{24}\ \mathrm{kg} \qquad m_M = 7.346 \times 10^{22}\ \mathrm{kg}$$

$$\eta = \frac{m_M}{m_E + m_M} \approx 0.012151 \qquad \frac{m_E}{m_M} \approx 81.30$$

- The rotating positions are $(-\eta, 0)$ and $(1-\eta, 0)$.
- The length unit is the constant separation in the circular model. A representative lunar scale is $a = 384{,}400\ \mathrm{km}$ [22, 23].
- The real lunar orbit is eccentric and perturbed by the Sun.

Five Earth–Moon Lagrange points



- L_1, L_2, L_3 lie on the Earth–Moon line; L_4, L_5 complete equilateral triangles [12].
- Bbarycenter approximates the earth's center. L4 leads in a ccw lunar orbit.

Collinear equilibria

- Set $y = 0$ and solve $\Omega_x = 0$.

$$x - (1-\eta)\frac{x+\eta}{|x+\eta|^3} - \eta\frac{x-1+\eta}{|x-1+\eta|^3} = 0$$

- L_1 lies between Earth and the Moon.
- L_2 lies beyond the Moon; L_3 lies beyond Earth on the opposite side.
- L_1, L_2 and L_3 have exponentially growing linear modes [13].
- Gravity supplies orbital acceleration. Gravity does not vanish at Lagrange points.

Triangular equilibria

1. For L_4 and L_5, both distances equal the primary separation.

$$\rho_1 = \rho_2 = 1$$

2. Equate the squared distances and solve for x.

$$(x+\eta)^2 = (x-1+\eta)^2 \quad \Longrightarrow \quad x = \frac{1}{2} - \eta$$

3. Substitute into $\rho_1^2 = 1$.

$$\boxed{L_4 = \left(\frac{1}{2} - \eta, \frac{\sqrt{3}}{2}\right) \qquad L_5 = \left(\frac{1}{2} - \eta, -\frac{\sqrt{3}}{2}\right)}$$

4. Both are 60° from the Moon as viewed from Earth [12].

Linear stability near L_4 and L_5

- Linearize about the the equilibrium $(x, y) = (x^*, y^*) + (\delta x, \delta y)$ to get

$$\delta x'' - 2\delta y' = \Omega_{xx}\delta x + \Omega_{xy}\delta y$$

$$\delta y'' + 2\delta x' = \Omega_{xy}\delta x + \Omega_{yy}\delta y$$

- Derivatives Ω_{xy} etc are evaluated at equilibria (x^*, y^*). The characteristic equation for a mode proportional to $e^{\lambda t}$ is

$$\boxed{\lambda^4 + \lambda^2 + \frac{27}{4}\eta(1-\eta) = 0}$$

- Purely imaginary, distinct eigenvalues require $27\eta(1-\eta) < 1$ [13].

Earth–Moon stability test

- For $0 < \eta \leq 1/2$, the stability condition is

$$\eta < \eta_{\mathrm{crit}} = \frac{1}{2}\left(1 - \sqrt{\frac{23}{27}}\right) \approx 0.038521$$

- The Earth–Moon mass fraction satisfies

$$\eta \approx 0.012151 < 0.038521 \qquad 27\eta(1-\eta) \approx 0.3241 < 1$$

- In the circular restricted model L_4 and L_5 are linearly stable and the remaining collinear points are unstable [13, 22, 23].

Meaning of stability

- Small disturbances near L_4, L_5 produce bounded oscillations in the linear model; they do not decay back to the equilibrium [13].
- Coriolis coupling is part of the stability mechanism. Position-dependent forces alone do not determine the linear modes.
- Linear stability describes sufficiently small disturbances in an ideal model. It does not certify every nearby nonlinear orbit.
- Lunar eccentricity, solar gravity, and radiation pressure change the spacecraft problem. Real operations need an orbit-specific assessment.

Earth–Moon and Sun–Earth systems

System	Geometry	Observatory examples
Earth–Moon	L_2 beyond the Moon; L_4, L_5 flank the lunar orbit	NCLE radio experiment at Earth–Moon L_2
Sun–Earth	L_1 sunward; L_2 about 1.5 million km anti-sunward	SOHO at L_1; Webb and Euclid around L_2

- Labels are different primary systems [20, 8, 14, 5] named explicitly in references.

Why observatories use Sun–Earth L_2

- Sun, Earth, and Moon remain on roughly the same side of the observatory, supporting shielding and a cold telescope [18].
- Orbits around L_2 provide a useful balance of thermal conditions, sky access, solar power, and Earth communication.
- Webb follows a large three-dimensional orbit around the L_2 region to balance avoiding eclipses, stray light and communication constraints [21].
- Science and engineering needs in addition to stability govern station choice.

Earth–Moon astronomy example

- The Netherlands–China Low-Frequency Explorer (NCLE) flew on the Queqiao mission to the Earth–Moon L_2 region in 2018 [20].
  - Not optical imaging, low-frequency radio experiment, with three antennas.
  - Science includes radio background, solar activity, and early-Universe radio signals.
  - Earth remains visible from NCLE's orbit.
- Earth–Moon L_2 supported the ARTEMIS probes in 2010. [17].

Stable points and telescope placement

- Earth–Moon L_4, L_5 offer linearly stable equilibria under the ideal model's assumptions [13]. Stable equilibrium does not automatically give a good thermal environment, viewing geometry, or communication links.
- Existing observatories use L_1 or L_2: NCLE at Earth–Moon L_2 and Webb at Sun–Earth L_2. [20, 14].
- None the less, an L_4 or L_5 Earth–Moon telescope has potential advantages.

Observatories in flight

Status checked 12 October 2026

Mission	Destination	Status and science
Webb	Sun–Earth L_2	Observing; infrared astronomy [14]
Euclid	Sun–Earth L_2	Observing; dark-Universe survey [5]
Roman	Sun–Earth L_2	Launched 30 August 2026. Early mission and commissioning with first images anticipated early 2027 [15]

- Launched observatory are not necessarily routine science ready.
- Gaia retired from its L_2 orbit in March 2025 [6].

Timeline of earlier observatories

Years are launch dates; locations refer to science operations

Year	Mission	Milestone
2001	WMAP	Sun–Earth L_2 microwave background mission; science ended in 2010 [19]
2009	Herschel	Sun–Earth L_2 infrared observatory; coolant exhausted in April 2013 [7]
2009	Planck	Sun–Earth L_2 microwave background mission; science ended in October 2013 [9]
2013	Gaia	Sun–Earth L_2 stellar mapping; spacecraft retired in March 2025 [6]

Timeline of recent milestones

Date	Milestone
2018	Queqiao carries NCLE to the Earth–Moon L_2 region [20]
Dec 2021 / Jan 2022	Webb launches on 25 December and reaches its Sun–Earth L_2 orbit on 24 January [14, 16]
July 2023	Euclid launches on 1 July for Sun–Earth L_2 [11]
March 2025	Gaia leaves the Sun–Earth L_2 region and is powered down [6]
August 2026	Roman launches on 30 August toward Sun–Earth L_2 [15]

Timeline of planned observatories

Agency targets checked 12 October 2026; schedules may change

Target	Mission	Location and purpose
March 2027	PLATO	Sun–Earth L_2 halo orbit; transiting exoplanets and stellar studies [10]
2031	Ariel	Sun–Earth L_2 halo orbit; exoplanet atmospheres [3]
2037	NewAthena	Sun–Earth L_1 region; X-ray observatory [2]

- These are planned launches, not confirmed operational arrival dates.
- Current mission pages supersede older factsheets showing earlier launch targets.

Station Keeping

- **Station Keeping**: Corrective maneuvers to keep a spacecraft near a desired orbit.
- Controllers estimate position and velocity, predict departures from target orbit, and program thruster to compensate for orbital instability and external disturbances.
- For a telescope near L_2, station keeping maintains the chosen orbit around the Lagrange point. The telescope continues moving along that orbit.

Source: STScI, JWST Orbit — Orbit maintenance

Station keeping for space telescopes

- **Station Keeping** uses orbit estimates and corrective burns to maintain the selected trajectory. Unstable L_1/L_2 regions require active orbit management [21].
- Webb's sunshield experiences radiation force and torque; reaction-wheel momentum unloading adds perturbations [21].
- Webb's nominal correction interval is about 42 days; negligible corrections may be skipped. Roman plans roughly 28-day intervals after orbit insertion [21, 1].
- Orbit correction and attitude control consume resources. Gaia's cold-gas depletion ended observing. Roman's early fuel savings increased its potential lifetime [4, 1].

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